

Samyak Kharat

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EDUCATION

MIT World Peace University <i>B.Tech in ECE AI&ML (4th Year)</i>	2024 – 2027 <i>Pune, MH</i>
MIT World Peace University <i>Diploma in ECE Ai&MI</i>	2021 – 2024 <i>Pune, MH</i>

WORK EXPERIENCE

Tech Team (MTT), ROBOCON <i>Robotics Control Developer</i>	2023 – 2026 <i>Pune, MH</i>
Lead — Head of the Bot Helped build the step-climbing mechanism and mecanum-drive autonomous navigation system for the team's DD Robocon 2026 robot, a contribution that helped the team finish All India Rank 2.	2026
Mentor — OpenCV Workshop Taught OpenCV and computer vision fundamentals through hands-on technical training for 1st- and 2nd-year students as part of the team's internal workshop series.	2025 – 2026
Senior Engineer - Developed the core control system for the jump bot, which the team used to land the season's basketball dunk shot. - Engineered a real-time auto-aim system using LiDAR, sensor fusion, and odometry, enabling closed-loop self-alignment and dynamic shooting-power adjustment.	2025
Engineer Built an autonomous ROS2 robot for ball sorting and storage, developing OpenCV-based navigation and object detection that let it locate and store balls without manual input.	2024
Trainee Engineer - Bridged hardware and software by integrating STM32 and ESP32 nodes over CAN bus and UART for the season's embedded control systems. - Designed and built a manually operated robot with an integrated embedded control system, laying the groundwork for the team's later automation efforts.	2023

ACHIEVEMENTS AND PROJECTS

DD Robocon (2023 Finalist, 2024 Semi-Finalist, 2025 Dunk Robot, 2026 Finalists) Achieved AIR 2 in DD Robocon 2026, AIR 3 in DD Robocon 2024, and AIR 2 in DD Robocon 2023 through competitive robotics design and autonomous systems development.

SKILLS

Programming Languages & Tools:

C, C++, Python, Linux, Git, Version Control

Robotics, Simulation & Automation:

ROS2, Gazebo, Rviz, ros2_control, Nav2, SLAM, Motion Planning, Autonomous Navigation, Sensor Fusion, Docker, Ansible, GitHub Actions, CI/CD, MuJoCo

Computer Vision:

OpenCV, Object Detection, Image Processing, Real-Time Vision Systems

Embedded Systems:

STM32, TM4C, ESP32, Embedded C, Microcontroller Programming, CAN Bus, UART, PWM, Real-Time Control Systems